KMT-2026-BLG-0521 / OGLE-2026-BLG-0058

Astrometry of the microlensing target — summary

Prepared 2026-09-12. Solution: ~/KMTdata/Results/v8/IFfinal_<field>.mat (variable IFsys), tie in Tie_<field>.mat.

1. The data and the algorithm

Data. Two KMTNet cut-outs of the same field, **BLG41** and **BLG01**, 300 x 300 pixels at 0.4 arcsec/pix, in I band, spanning 2016 to 2026 in ten observing seasons (2020 is a short stub and is dropped). **19 133** and **19 981** exposures respectively, taken alternately on the same nights, a median of 12.2 minutes apart. The target sits at the field centre at I_{OGLE} = 18.13, RA 17:52:38.09, Dec –31:47:36.1.

Calibration set. Stars are selected on magnitude and isolation: **14 < I < 19** with **no companion brighter than I = 18 within 2 arcsec**, the companion list taken from the OGLE catalogue, which resolves about six times more sources than KMT detects. The set is then **iterated once**: a first solution is computed, and stars whose residual RMS lies more than **2 sigma** above the running median for their magnitude are dropped and the fit repeated. The cut is made in log space, so it is multiplicative and does not simply shave off the faint end. This leaves **260** stars on BLG41 and **235** on BLG01. The target passes the selection on its own merits. Everything else is dropped at the input, so all measurement, detrending and quality cuts act on this set alone.

The fit. Each exposure gets a full 2-D affine transformation, 6 parameters, and each star a position and proper motion, 4 parameters; these are solved together by alternating least squares. Alongside them the fit removes differential chromatic refraction (fitted in 6 equal-population colour bins), an annual term, a pixel-phase term, and two SysRem components computed once over the whole decade. Every star is freely fitted — nothing is held at a catalogue value — and each is weighted by its own residual scatter, with moving-median outlier rejection.

Absolute frame. A free fit determines only *relative* astrometry, so the result is tied to Gaia DR3 afterwards. The tie is fitted on stars with **RUWE < 1.4 and 15 < I < 17** and removes the full six-parameter gauge — a constant *and* a linear gradient across the field — then is applied to every source.

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2. Results

Achieved accuracy, residual RMS over the decade after position and proper motion are removed:

	BLG41 ΔX / ΔY	BLG01 ΔX / ΔY
calibration stars brighter than I = 17	**4.80 / 4.83 mas**	**4.75 / 4.95 mas**
the target, I = 18.13	**23.77 / 19.67 mas**	**21.51 / 20.45 mas**

The frame is therefore good to under **5 mas** on well-measured stars, and the target — three magnitudes fainter and blended — is measured to about **20 mas** per epoch, consistent with the running median of the residual RMS at its magnitude.

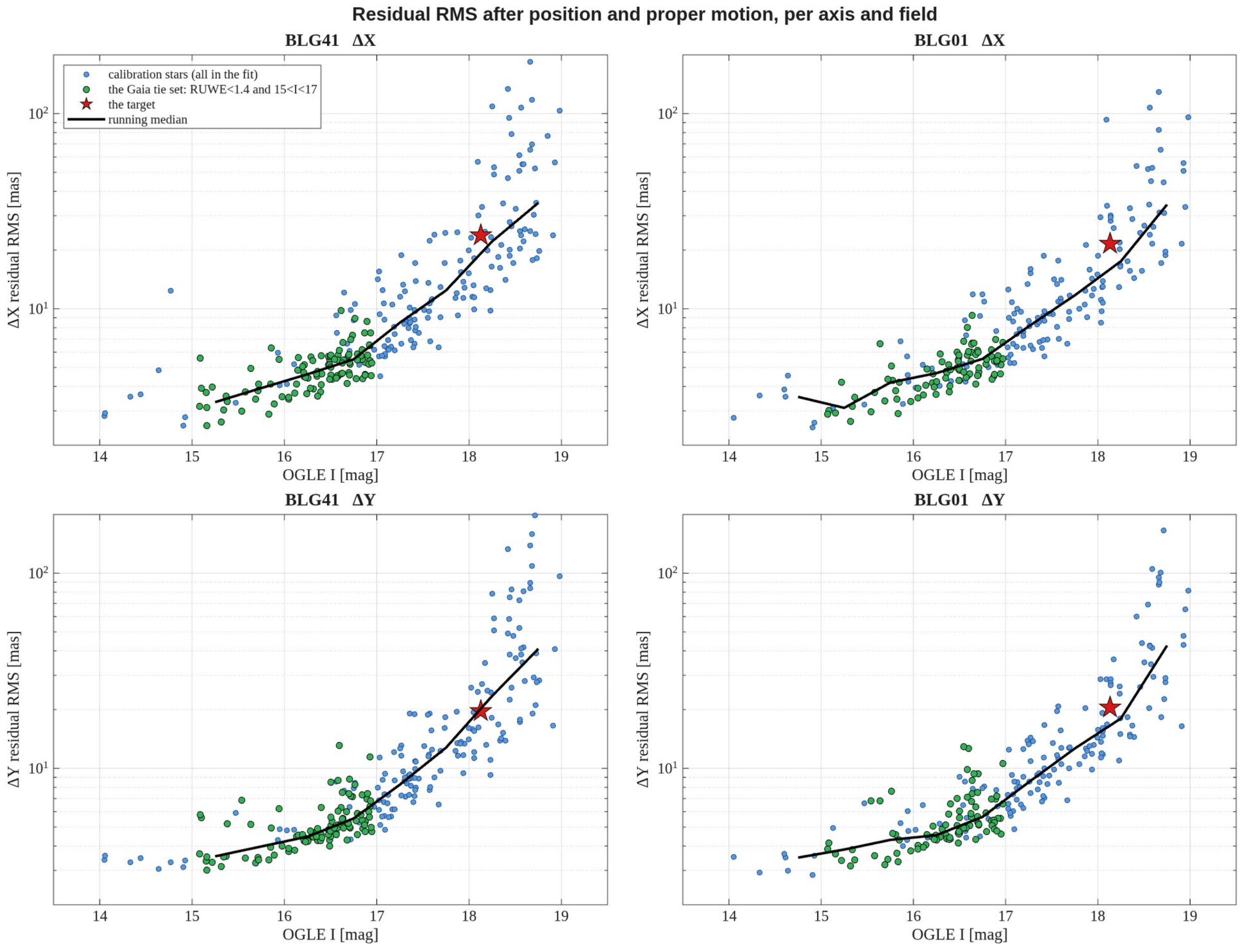
Proper motion of the target, absolute, after the Gaia tie:

	$\mu_\alpha \cos\delta$	μ_δ
BLG41	**–1.966 ± 0.347**	**–7.019 ± 0.358**

BLG01	**−1.514 ± 0.227**	**−7.073 ± 0.376**
difference between the fields	−0.452 ± 0.415	+0.054 ± 0.519

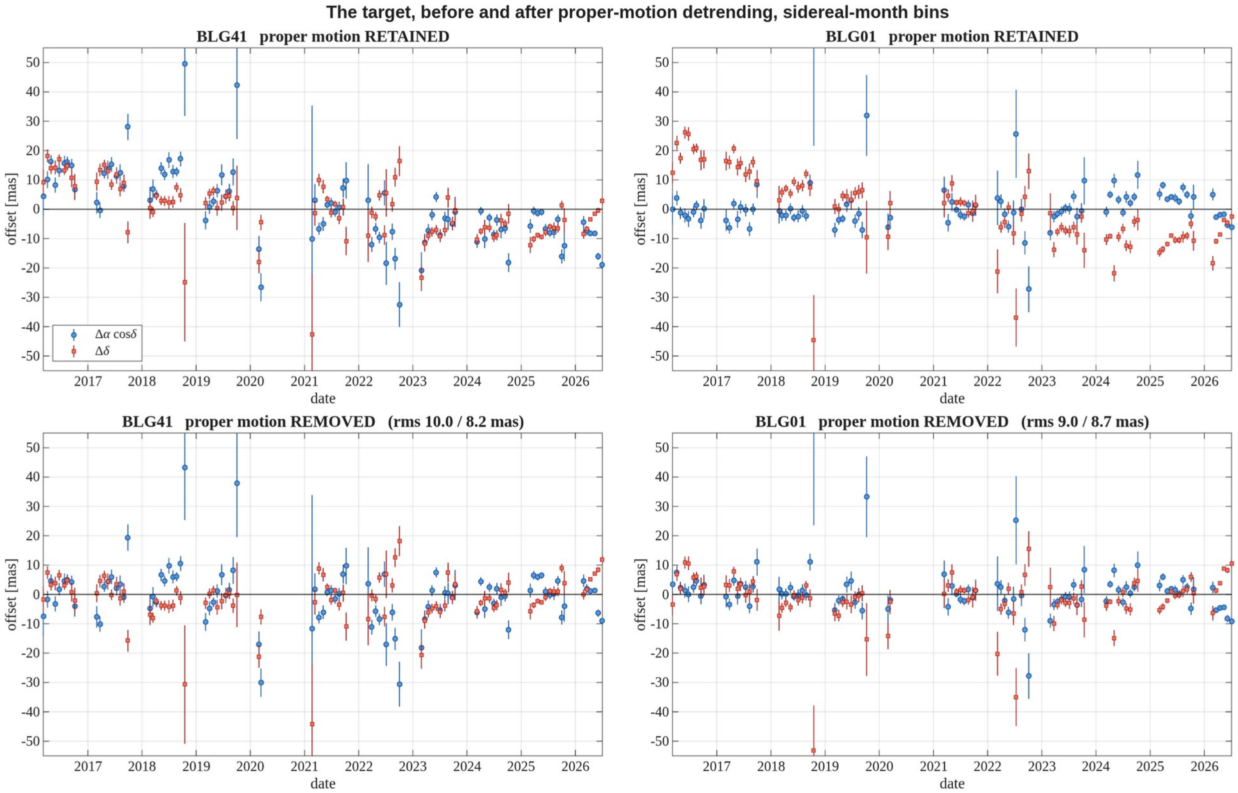
mas/yr. The two independent fields agree within their errors, at 1.09 and 0.10 sigma. The quoted error is internal, from the scatter of the ten season means about the fitted line; the external check against Gaia gives about **1.0 / 0.62 mas/yr** per star, and that is the figure to use in practice.

No astrometric anomaly is detected at the target. Its χ^2 on monthly binned residuals is 1500 and 1313 for 180 degrees of freedom, against field medians of χ^2/DoF of 6.5 and 6.8 — the target sits inside the body of the distribution, not in the tail. Whatever astrometric excursion the microlensing event produces is smaller than the scatter of an ordinary star of this brightness.



Residual RMS against the OGLE catalogue magnitude, per axis and per field, after position and proper motion have been removed. Blue: all 260 / 235 calibration stars, every one freely fitted. Green: the stars the Gaia tie is fitted on, RUWE below 1.4 and 15 < I < 17. Red: the target. Black: running median. The dashed lines mark the 14 and 19 mag selection limits. Compared with a set that has not been iterated, the 2 sigma cut removes the scattered points that previously sat well above the median relation.

`report_v8_RMS.png`



The target before and after proper-motion detrending, in both fields. Columns are the fields; the top row keeps the proper motion and the bottom row removes it. Blue circles are the offset in right ascension, red squares in declination, binned in one sidereal month with the standard error within the bin. The proper motion is plainly visible as the declination trend in the top row and is absent from the bottom row; the residual RMS quoted in the lower panels is per bin.

`report_v8_target_beforeafter.png`

More details

3. Object selection, step by step

step	BLG41	BLG01
raw sources in the MSc file	1177	1187
survive the standard quality cuts	594	621
have an OGLE I magnitude	540	478
in 14 < I < 19	529	470
... and isolated: no I < 18 companion within 2.0 arcsec	287	253
... and surviving the 2 sigma RMS cut	**260**	**235**

The isolation cut is the dominant one, removing 46% of the magnitude-selected sources on both fields. The outlier iteration then removes a further **27 and 18** stars — 9% and 7% — spread across all magnitudes rather than concentrated at the faint end: 6%, 14%, 10% and 3% by magnitude band on BLG41. Their median residual RMS is 22.4 and 48.5 mas against 10.8 and 10.6 mas for the stars kept.

For the tie only, Gaia is required: 247 of 260 and 207 of 235 sources have a counterpart, of which 198 and 163 pass RUWE < 1.4, and 104 and 85 also fall in 15 < I < 17.

4. The Gaia tie

The map. Gaia standard coordinates are fitted directly onto cut-out pixels from the tie stars — not carried through OGLE, which would inherit its ~110 mas astrometric error. Median residual **22.1 mas (BLG41)** and **21.8 (BLG01)**, with scales of 2.5195 and 2.5289 pix/arcsec against the 2.500 expected at 0.4″/pix.

The gauge. In the proper motions the gauge freedom is a constant plus a linear gradient across the field — six parameters. Removing only the constant leaves a shear of about −0.036 mas/yr per pixel, reproduced at nearly the same value in both independently reduced fields. With the full gauge removed, the scatter of our proper motions about Gaia is **0.994** / **0.608 mas/yr** (BLG41) and **1.049** / **0.641** (BLG01); removing only the constant would leave 3 to 4 times that. The gauge constant removed is −0.126 / +5.105 and +2.151 / +4.324 mas/yr.

5. What the fit does, in detail

Each of the two steps runs **two passes** internally. `IFsys.AnnualEffect` and `IFsys.PixPhase` read `0` in the saved solutions, which does *not* mean those effects were ignored: they are fitted and subtracted from the data in pass 1 and switched off in pass 2, which works on data from which they have been removed.

	pass 1	**between**	**pass 2 (final)**
per-epoch affine	fitted		fitted
per-source position and motion	fitted		fitted
DCR / chromatic refraction	fitted		fitted again
annual term	fitted and subtracted		forced off
pixel-phase term	fitted and subtracted		forced off
SysRem		applied	
iterative reweighting	15 iterations		6 iterations

- **Per-epoch affine** — 6 parameters per exposure in `ParE`, stored as the deviation from the identity: translation (2), rotation (1), uniform scale (1), shear (2). 520 equations for 6 unknowns each epoch. Translation dominates at about 37 / 28 mas rms; rotation and scale each move a star at the cut-out edge by roughly 10 mas.
- **Per source** — position at `JD0` = 2019-06-01 and proper motion, 4 parameters in `ParS`, in pixels and pixels per year at 400 mas/pix.
- **DCR** — `[1, sin(pa)·secz, cos(pa)·secz]` plus six higher orders in $(\mathtt{secz·sin\ pa})^n$, $(\mathtt{secz·cos\ pa})^n$ for n = 2, 3, 4: 18 parameters per colour bin, 6 equal-population bins. Only the *differential* part is recoverable — refraction common to the field is exactly a translation and is absorbed by `t_x, t_y` every epoch.
- **Annual term** — a quartic in the phase of the calendar year, 10 parameters, fitted globally, one curve shared by every star. Not parallax, which is per-source and switched off.
- **Pixel phase** — a quintic in the sub-pixel phase, 10 parameters, global, recoverable only because the pipeline stored the per-epoch registration shifts.
- **SysRem** — 2 components over the whole decade, reused rather than refitted per season, with point-wise rejection of bad cells. Refitting per season was tested and is worse: see section 8.

Timings and convergence:

	BLG41	**BLG01**
step 1, decade SysRem	16.9 min	15.4 min

step 2, final solution	31.1 min	29.1 min
convergence, last 3 iterations	0.0002 mas	0.0047 mas

No per-season fitting enters the solution; seasons are used only for reporting.

6. Comparison stars

Six, chosen to be the same physical objects in both fields — three near the target's magnitude and three brighter. All six survive the outlier cut and sit at identical positions in the two fields:

tag	I	dist [pix]	decade RMS ΔX / ΔY, BLG41	BLG01
`m18_d04`	18.14	4.5	28.86 / 25.64	25.24 / 27.09
`m18_d17`	18.09	17.3	44.90 / 23.22	80.36 / 27.53
`m18_d48`	18.05	48.4	18.06 / 15.57	13.86 / 15.58
`m16_d24`	16.68	24.4	**4.14** / **4.31**	**4.13** / **4.32**
`m16_d30`	16.63	30.4	6.69 / 5.53	6.55 / 5.62
`m16_d31`	16.75	30.6	8.42 / 7.82	11.16 / 7.84

mas. The bright trio reproduces between the fields almost exactly — `m16_d24` gives –3.22 / –7.85 on BLG41 and –3.22 / –7.85 on BLG01 — while the faint trio does not: `m18_d17` reaches 45 and 80 mas decade RMS on only ~2400 usable epochs. It is a poor star.

`m18_d04` is the neighbour blended with the target, 4.5 pixels away under ~7 pixel seeing. It is a legitimate member of the calibration set — the target at I = 18.13 is fainter than the I < 18 companion threshold, so it does not trigger rejection — but its centroid tracks the target's brightening through the event, and it is not an independent object.

Coordinates. Pixel positions are per field, since the cut-outs are offset by about 1.1 pixels; sky positions come from inverting each field's own map and the two fields agree on them to 2 to 9 mas.

star	I	BLG41 pixel	BLG01 pixel	RA (J2000)	Dec (J2000)
target	18.13	150.4, 150.4	151.6, 150.2	17:52:38.085	–31:47:36.21
`m18_d04`	18.14	146.5, 148.3	147.6, 148.0	17:52:38.210	–31:47:37.06
`m18_d17`	18.09	134.5, 143.6	135.6, 143.4	17:52:38.590	–31:47:39.00
`m18_d48`	18.05	198.6, 145.4	199.8, 145.0	17:52:36.585	–31:47:38.10
`m16_d24`	16.68	161.3, 172.2	162.4, 172.1	17:52:37.746	–31:47:27.53
`m16_d30`	16.63	120.4, 154.6	121.4, 154.5	17:52:39.023	–31:47:34.60
`m16_d31`	16.75	138.7, 178.7	139.8, 178.6	17:52:38.436	–31:47:24.98

7. Chi-squared of the monthly binned residuals

For every star the residuals are binned in sidereal months; each bin contributes `(mean/standard error)²` in both axes, and `DoF = 2·N_bins − 4`. Values are not divided by DoF.

	BLG41	BLG01

stars measured	260	235
median DoF	182	182
target χ^2	**1500** (DoF 180)	**1313** (DoF 180)
target χ^2/DoF	8.3	7.3
field median χ^2/DoF	6.5	6.8

The target is unremarkable. The field median of χ^2/DoF near 7 is itself informative: the standard error of the mean within a month **understates the true bin uncertainty by about $\sqrt{7} \approx 2.6$**, because the noise within a month is correlated rather than independent. The effect is multiplicative and nearly independent of magnitude, and the faintest stars come closest to unity, since for them genuine photon noise — which *is* independent between exposures — dominates. The error bars of the RA/Dec and motion figures are the standard error scaled by this factor, computed per star and per axis and quoted in the panel titles.

8. A residual systematic that survives every correction

Fitting a straight line to each star's residuals **within each season separately** leaves slopes that are real and unexplained.

	BLG41	BLG01
median per-star season-slope rms	2.90 / 2.80 mas	2.93 / 2.90 mas
expected from white noise	0.99 mas	0.88 mas
excess over white noise	**2.9x**	**3.3x**

They are **not noise**. Matching 181 stars between the fields through OGLE and correlating each star's ten-season slope pattern gives a median correlation of **+0.795** in one axis and **+0.754** in the other, with **96 to 97% of stars positive**. Two independent reductions, independent frames, essentially every star showing the same season-by-season pattern: about 79% of the slope variance is shared systematic rather than measurement noise. The target is entirely typical in this respect, at the 51st to 75th percentile among stars within 0.4 mag of it.

The cause has not been identified. Seven candidates were tested and eliminated:

candidate	test
DCR / colour	per-season slope against colour
DCR colour resolution	full refit with 12 and with 20 colour bins instead of 6
calibrator contamination	second clipping iteration on the season scale, refit without the 21 + 22 stars with 2σ-high season-offset χ^2
blending	against nearest-neighbour distance
field distortion	spatial correlation; quadratic in position
pixel phase	slope against within-season phase drift
SysRem cadence	refit per season instead of per decade

The colour-bin refits are the direct test of the DCR model. The 18 DCR parameters are fitted independently in each colour bin, so with 12 or 20 bins (19 or 11 stars per bin instead of 37) any colour dependence too fine for six bins would have been absorbed. Nothing changed:

	6 bins (v8)	12 bins	20 bins
slope rms, BLG41	2.90 / 2.80	2.98 / 2.77	2.92 / 2.80
slope rms, BLG01	2.93 / 2.90	2.97 / 2.89	2.92 / 2.83
cross-field correlation	+0.795 / +0.754	+0.794 / +0.742	+0.796 / +0.766
bright-star RMS, BLG41	4.80 / 4.83	4.74 / 4.83	4.74 / 4.83
bright-star RMS, BLG01	4.75 / 4.95	4.78 / 4.95	4.73 / 4.96
target RMS, BLG41	23.77 / 19.67	23.80 / 19.72	23.78 / 19.66
target RMS, BLG01	21.51 / 20.45	21.47 / 20.40	21.53 / 20.34
target PM, BLG41	−1.966 / −7.019	−1.879 / −7.020	−1.909 / −7.116
target PM, BLG01	−1.514 / −7.073	−1.684 / −7.048	−1.595 / −7.105

All in mas or mas/yr, X / Y. The cross-field correlation, which is the quantity that measures the shared systematic, is unchanged to the third decimal, the bright-star RMS moves by at most 0.06 mas and the target's proper motion by at most 0.17 mas/yr, or 0.75 of its error. Six colour bins already capture all the colour dependence there is; the six-bin solution is kept.

The season-scale clipping asks whether the frame is being pulled by calibrators that themselves carry coherent season offsets — the decade RMS used for the v8 cut averages such offsets away (a 16.75-mag star among the comparison set has its X residual at +15 mas for two seasons and −15 mas for one, with a perfectly normal decade RMS). Applying a second 2σ cut, in log space per magnitude bin, on each star's season-offset χ^2/DoF — Σ(season mean / s.e.)^2 over seasons and axes — removes 21 stars in BLG41 and 22 in BLG01, two-thirds of them at 16–17 mag; the target sits at z = +0.9 / +0.6, inside the population. Refitting on the 239 + 213 survivors changes nothing on the stars common to both solutions:

common stars	BLG41	BLG01
RMS, I < 17 (117 / 105 stars)	4.66 / 4.77 → 4.65 / 4.80	4.61 / 4.95 → 4.60 / 4.94
RMS, 17–18	+1.0% / +0.5%	−1.5% / +1.5%
RMS, 18–19	+0.1% / +2.4%	−0.9% / +1.1%
Gaia scatter, RUWE-clean (181 / 146)	0.967 / 0.558 → 0.934 / 0.572	1.067 / 0.629 → 0.982 / 0.661
target PM	−1.966 / −7.019 → −2.003 / −6.997	−1.514 / −7.073 → −1.770 / −7.024

Per-star RMS moves by under 1% with half the stars going each way; the Gaia scatter improves in X and worsens in Y; the target's proper motion moves by 0.04 and 0.26 mas/yr (0.1σ and 1.1σ), bringing the two fields to within 0.23 mas/yr in X. The apparent 3% gain in the quoted bright-star RMS (4.64 / 4.72 and 4.58 / 4.94) is entirely the removal of the worst stars from the median, not an improvement of the solution. The per-epoch frame with some 240 stars is insensitive to these 8–9%, so calibrator contamination is not the origin of the slopes, and v8 is kept as the reference.

The per-season SysRem test is the most informative. SysRem represents residuals as a per-source coefficient times a per-epoch mode, which is exactly the structure the slopes appear to have; giving it its own two components in each season — ten times the freedom — should have absorbed them. Instead the slopes grew and every other metric degraded by 5 to 11%, because each season's correction is constrained by about 1900 epochs instead of 19 000 and only 88 to 90% of epochs received any correction at all. The slopes are therefore **not** a low-rank common mode within seasons, consistent with their being per-star and spatially uncorrelated.

Since the effect reproduces between two cut-outs of the same stars on the same nights, it is in the images rather than in the reduction. It is the likely origin of the χ^2/DoF of about 7 in section 7, of the factor 2.6 by which the internal proper-motion errors are optimistic, and of the floor on what this dataset can deliver.

9. Overlap of the two fields

Stars are identified through the OGLE catalogue; matching the two source lists directly is ambiguous because a close pair may be split in one field and merged in the other.

	BLG41	BLG01
sources	594	621
matched to an OGLE entry	551 (93%)	497 (80%)
distinct OGLE stars behind them	**454**	**394**

The gap between the last two rows is the blending, measured: 551 detections correspond to only 454 stars on BLG41, so about 97 detections are duplicates.

OGLE I	BLG41	BLG01	in both	distinct	found in both
< 15	28	22	22	28	78.6%
15 – 16	54	42	42	54	77.8%
16 – 17	172	149	146	175	83.4%
17 – 18	135	119	117	137	85.4%
18 – 19	55	44	41	58	70.7%
all brighter than 19	**444**	**376**	**368**	**452**	**81.4%**

About four stars in five are detected in both fields, flat from the brightest down to I = 18. **There is nothing fainter than I = 19 to compare**: our source lists span 12.87 to 18.98 in both fields, while OGLE itself reaches 21.40 and holds 2660 entries fainter than 19. That is the KMT detection limit in this crowding, not a catalogue cut.

10. Files

file	content
`report_v8_RMS.png`	residual RMS against OGLE I, per axis and field
`report_v8_target_beforeafter.png`	the target before and after proper-motion detrending
`report_v8_pm_vs_gaia.png`	our absolute proper motion against Gaia's, after the tie
`report_v8_chi2.png`	χ^2 distribution of the monthly binned residuals, target marked

`report_v8_radec_<field>_<tag>.png`	RA and Dec residuals against time and against each other, colour-coded by time; bi dropped
`report_v8_motion_<field>_<tag>.png`	2x2 motion figures, proper motion retained and removed; same binning cuts as the RA/
`source_motion_v8_<field>[_<tag>].csv`	per-epoch positions for the target and each comparison star

CSV columns are `JD, X_mas, Y_mas, errX_decade, errY_decade, errX_season, errY_season, season`, with positions relative to that star's own mean and **proper motion not removed**. Headers carry the star's magnitude, pixel position, sky position and absolute proper motion.

In the motion figures the columns are the two axes, **X left and Y right**; the top row keeps the proper motion and the bottom row removes it. The legend quotes the motion along the pixel axis and the heading quotes it on the sky. **Pixel X runs opposite to RA**, so those two carry opposite signs in X.

Figures

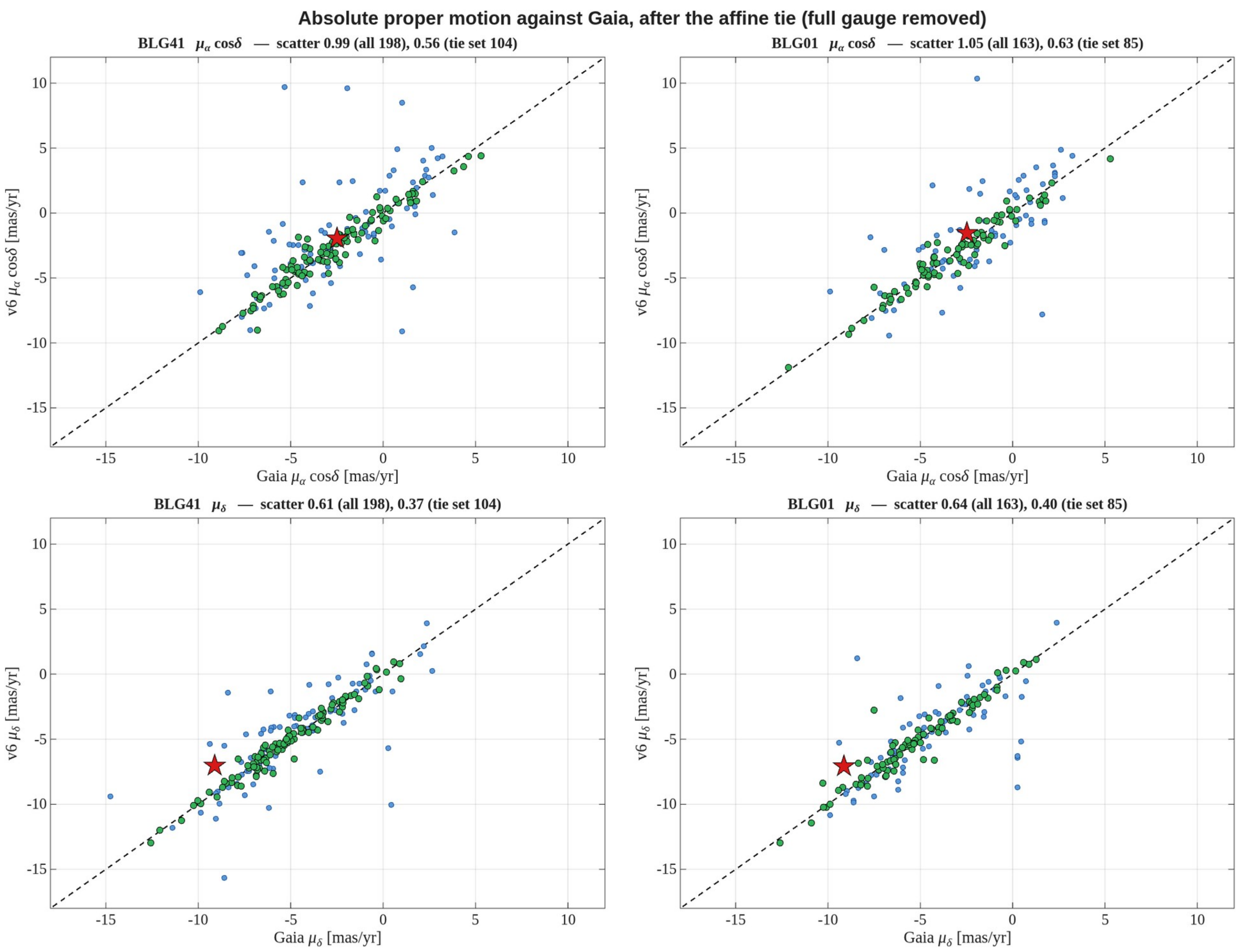


Figure 1. Our absolute proper motion against Gaia's, per axis and per field, after the affine tie with the **full six-parameter gauge** removed. Dashed line is 1:1. The scatter quoted in each panel is the robust standard deviation of the difference; it improves by a factor of three to four over a tie that removes only the constant. The target is the red star.
`report_v8_pm_vs_gaia.png`

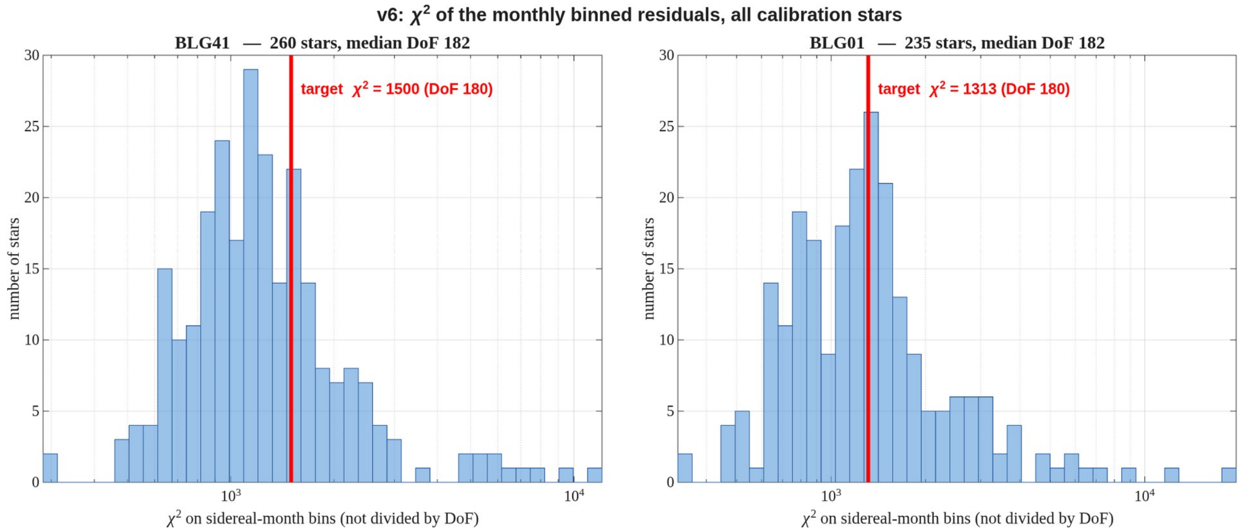


Figure 2. χ^2 of the sidereal-month binned residuals for every calibration star, both axes combined, **not divided by DoF**. Each bin contributes (mean / standard error)² per axis and DoF = $2N_{\text{bins}} - 4$. The red line marks the target, which sits in the body of the distribution rather than the tail.

`report_v8_chi2.png`

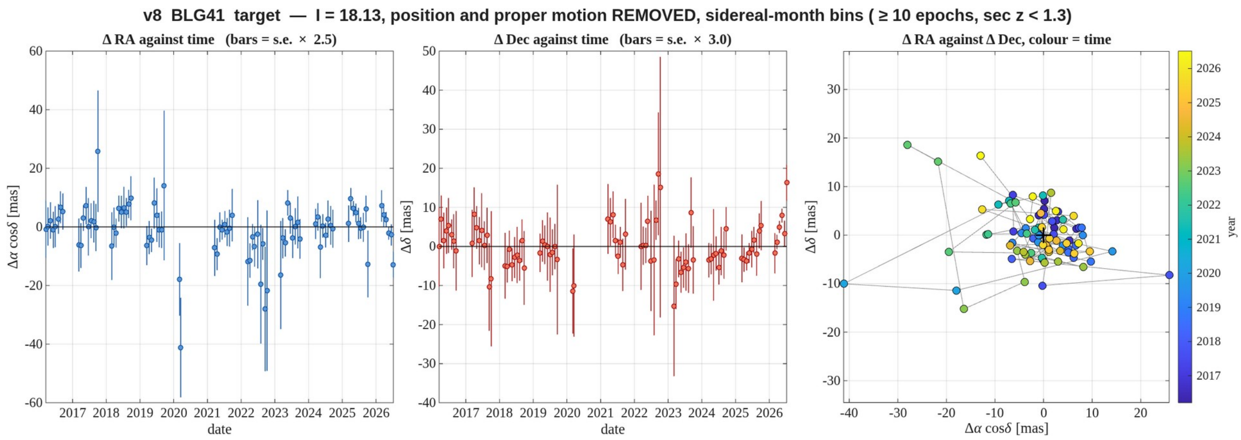


Figure 3. **BLG41** — the target, I = 18.13. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour; consecutive months are joined by a thin grey line. Sidereal-month bins; error bars are the standard error within the bin scaled by the empirical factor $\sqrt{\chi^2/\text{DoF}}$ of the bin means, quoted in each panel title, so that they carry the real bin-to-bin scatter (section 7). Bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped: the season-edge months, with a handful of high-airmass frames, otherwise dominate the plot.

`report_v8_radec_BLG41_target.png`

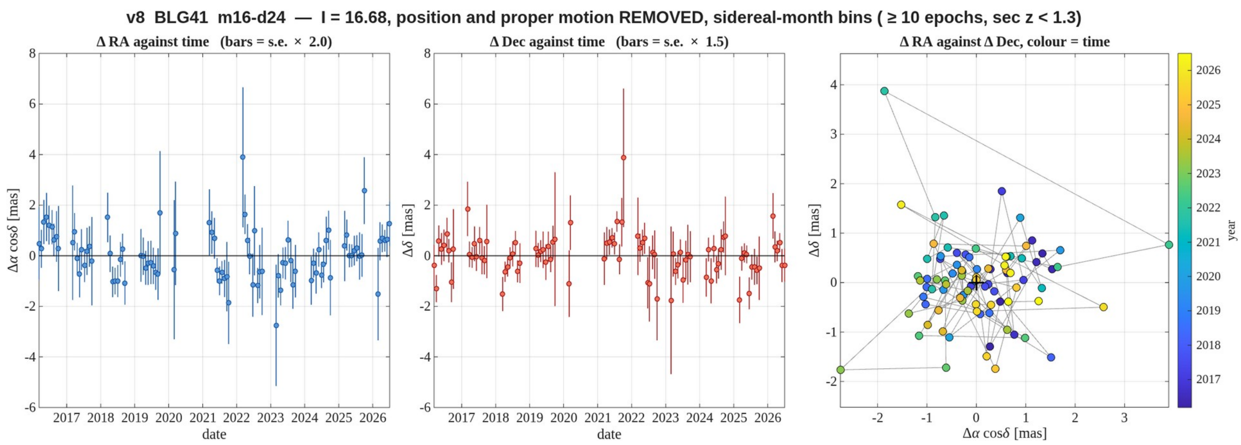


Figure 4. **BLG41** — comparison star I = 16.68, 24 pix from the target — the best measured of the set. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour; consecutive months are joined by a thin grey line. Sidereal-month bins; error bars are the standard error within the bin scaled by the empirical factor √(χ²/DoF) of the bin means, quoted in each panel title, so that they carry the real bin-to-bin scatter (section 7). Bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped: the season-edge months, with a handful of high-airmass frames, otherwise dominate the plot.

`report_v8_radec_BLG41_m16_d24.png`

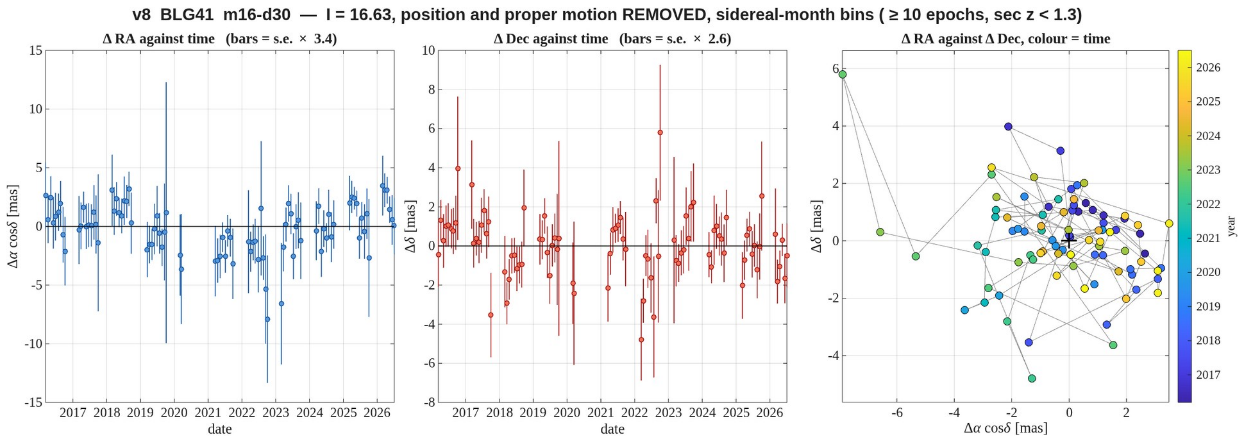


Figure 5. **BLG41** — comparison star I = 16.63, 30 pix from the target. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour; consecutive months are joined by a thin grey line. Sidereal-month bins; error bars are the standard error within the bin scaled by the empirical factor √(χ²/DoF) of the bin means, quoted in each panel title, so that they carry the real bin-to-bin scatter (section 7). Bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped: the season-edge months, with a handful of high-airmass frames, otherwise dominate the plot.

`report_v8_radec_BLG41_m16_d30.png`

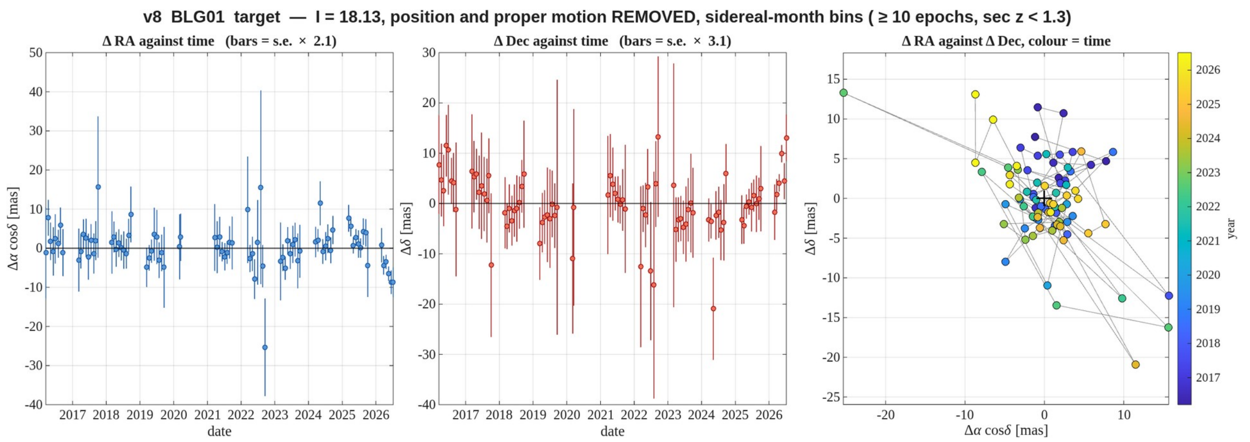


Figure 6. **BLG01** — the target, I = 18.13. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour; consecutive months are joined by a thin grey line. Sidereal-month bins; error bars are the standard error within the bin scaled by the empirical factor √(χ²/DoF) of the bin means, quoted in each panel title, so that they carry the real bin-to-bin scatter (section 7). Bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped: the season-edge months, with a handful of high-airmass frames, otherwise dominate the plot.

`report_v8_radec_BLG01_target.png`

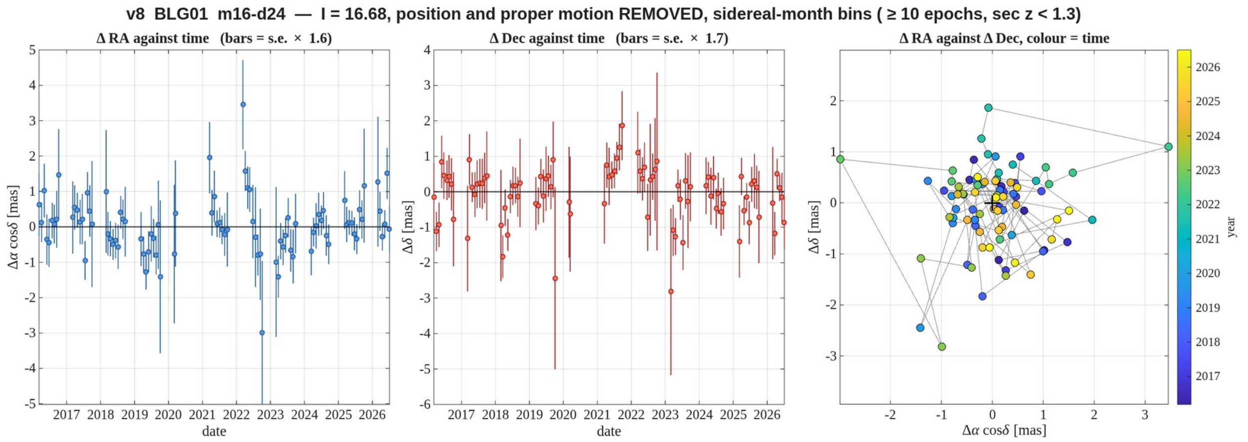


Figure 7. **BLG01** — comparison star I = 16.68, 24 pix from the target — the best measured of the set. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour; consecutive months are joined by a thin grey line. Sidereal-month bins; error bars are the standard error within the bin scaled by the empirical factor √(χ²/DoF) of the bin means, quoted in each panel title, so that they carry the real bin-to-bin scatter (section 7). Bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped: the season-edge months, with a handful of high-airmass frames, otherwise dominate the plot.

`report_v8_radec_BLG01_m16_d24.png`

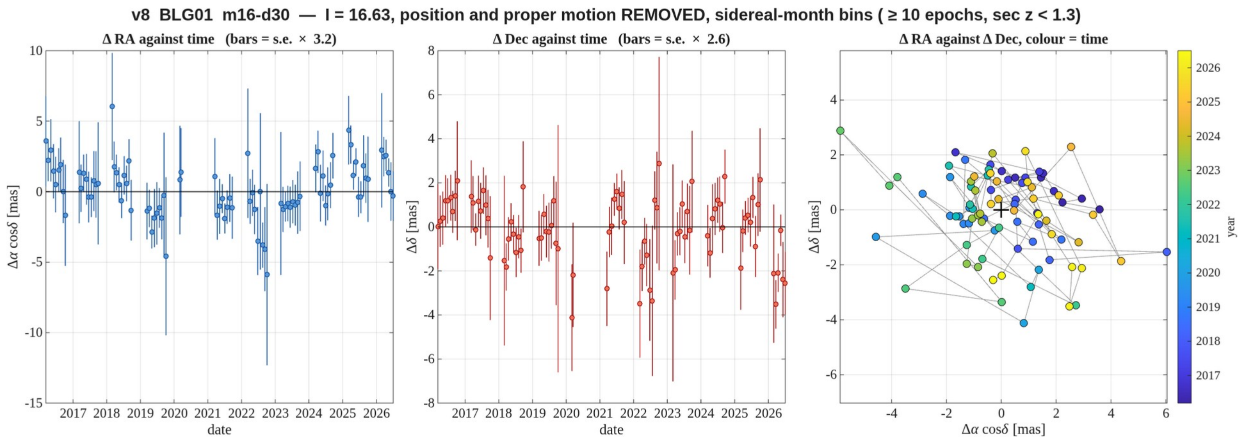


Figure 8. **BLG01** — comparison star I = 16.63, 30 pix from the target. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour; consecutive months are joined by a thin grey line. Sidereal-month bins; error bars are the standard error within the bin scaled by the empirical factor √(χ²/DoF) of the bin means, quoted in each panel title, so that they carry the real bin-to-bin scatter (section 7). Bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped: the season-edge months, with a handful of high-airmass frames, otherwise dominate the plot.

`report_v8_radec_BLG01_m16_d30.png`

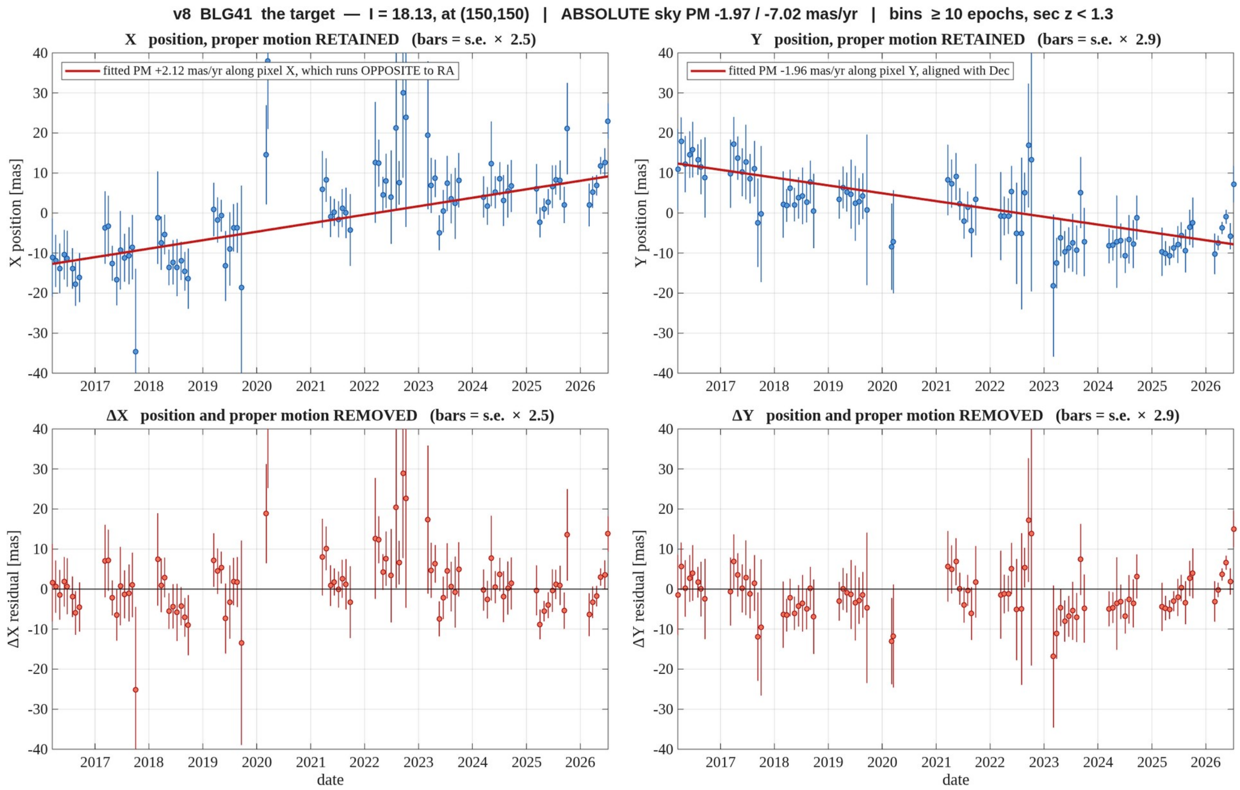


Figure 9. **BLG41** — the target, I = 18.13. Binned in one sidereal month, error bars are the standard error within the bin scaled by the empirical factor $\sqrt{\chi^2/\mathrm{DoF}}$ of the bin means (quoted in the panel titles, computed on the proper-motion-removed residuals and applied to both rows); bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.
`report_v8_motion_BLG41_target.png`

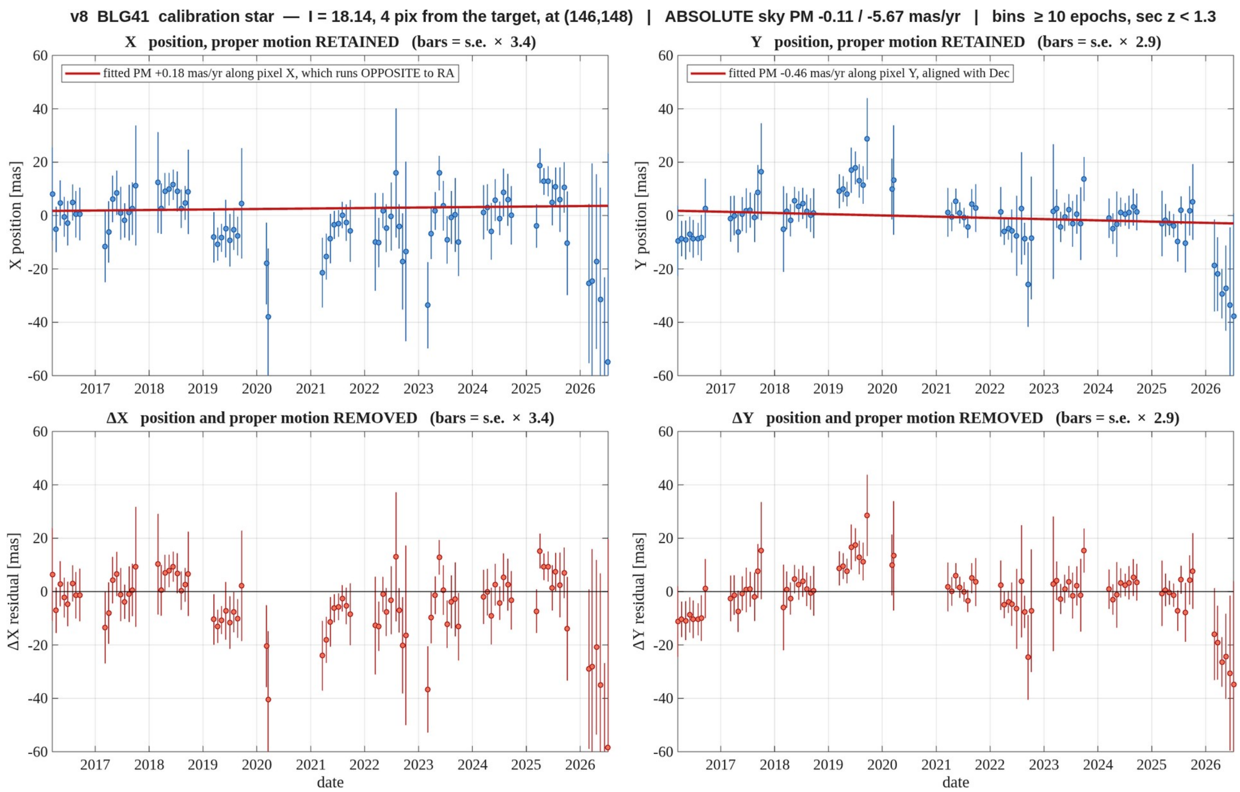


Figure 10. **BLG41** — I = 18.14, 4.5 pix away — **blended with the target**, see section 7. Binned in one sidereal month, error bars are the standard error within the bin scaled by the empirical factor √(χ²/DoF) of the bin means (quoted in the panel titles, computed on the proper-motion-removed residuals and applied to both rows); bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.
`report_v8_motion_BLG41_m18_d04.png`

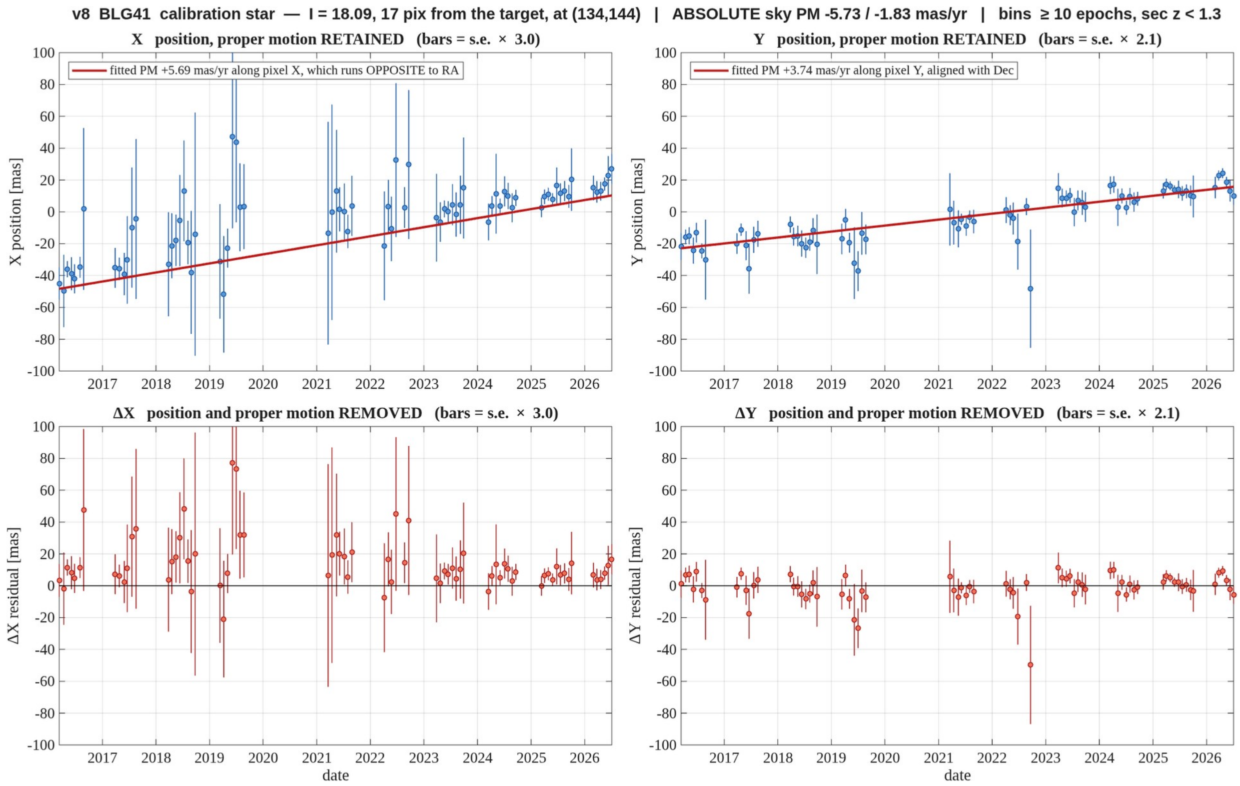


Figure 11. **BLG41** — I = 18.09, 17 pix away — poorly measured, see section 7. Binned in one sidereal month, error bars are the standard error within the bin scaled by the empirical factor √(χ²/DoF) of the bin means (quoted in the panel titles, computed on the proper-motion-removed residuals and applied to both rows); bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.
`report_v8_motion_BLG41_m18_d17.png`

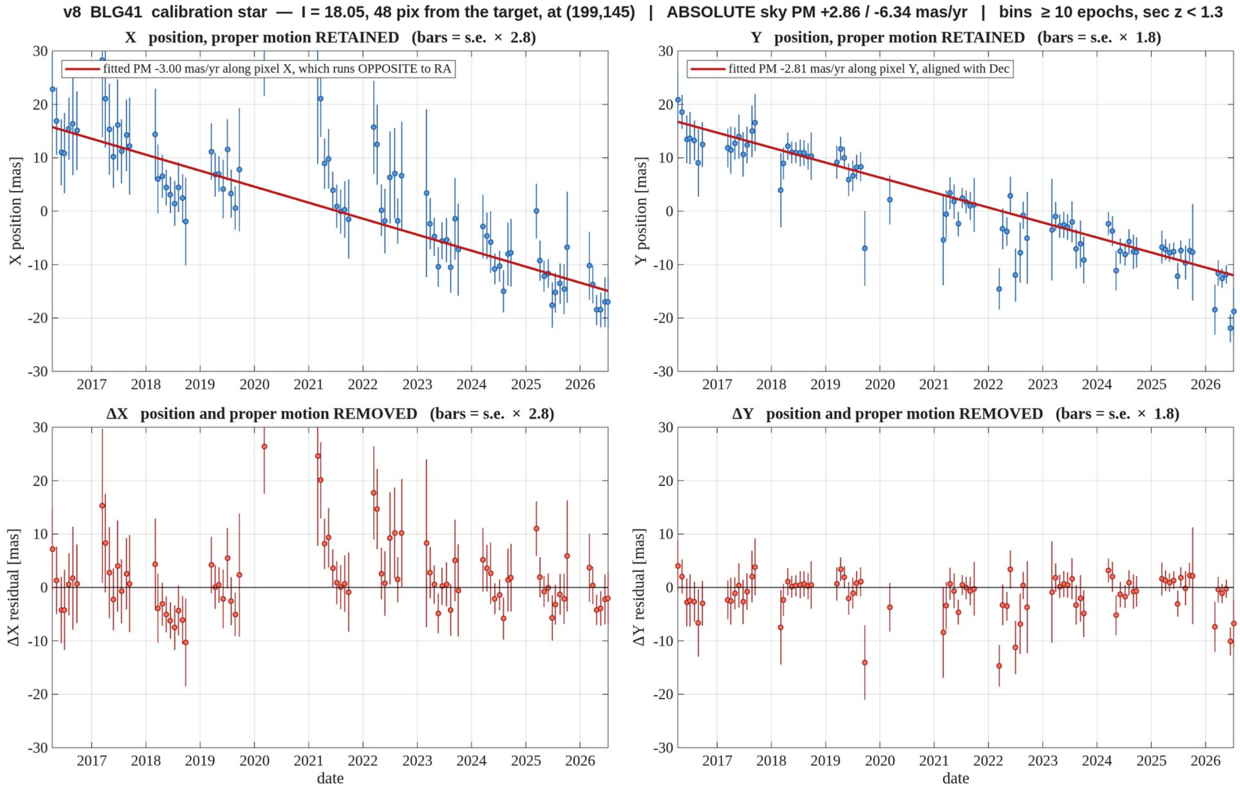


Figure 12. **BLG41** — I = 18.05, 48 pix away. Binned in one sidereal month, error bars are the standard error within the bin scaled by the empirical factor $\sqrt{(\chi^2/\mathrm{DoF})}$ of the bin means (quoted in the panel titles, computed on the proper-motion-removed residuals and applied to both rows); bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v8_motion_BLG41_m18_d48.png`

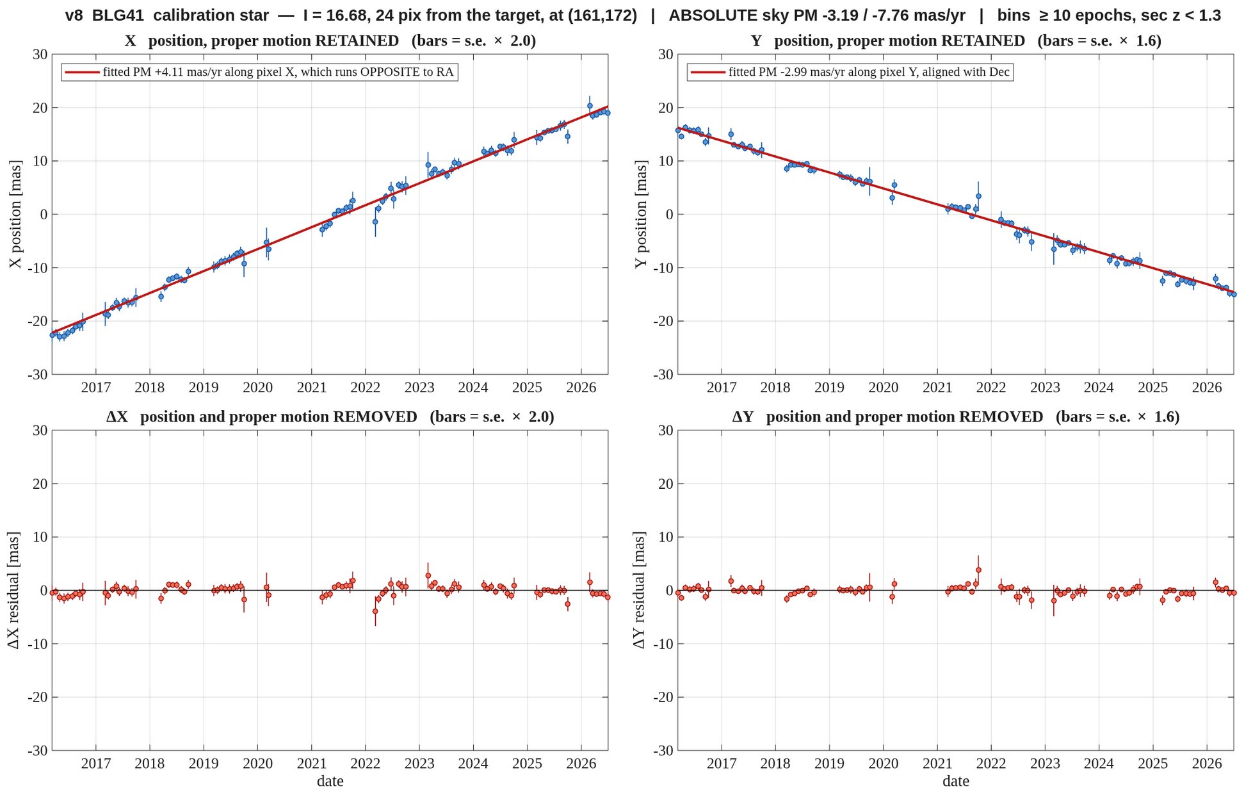


Figure 13. **BLG41** — I = 16.68, 24 pix away. Binned in one sidereal month, error bars are the standard error within the bin scaled by the empirical factor √(χ²/DoF) of the bin means (quoted in the panel titles, computed on the proper-motion-removed residuals and applied to both rows); bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.
`report_v8_motion_BLG41_m16_d24.png`

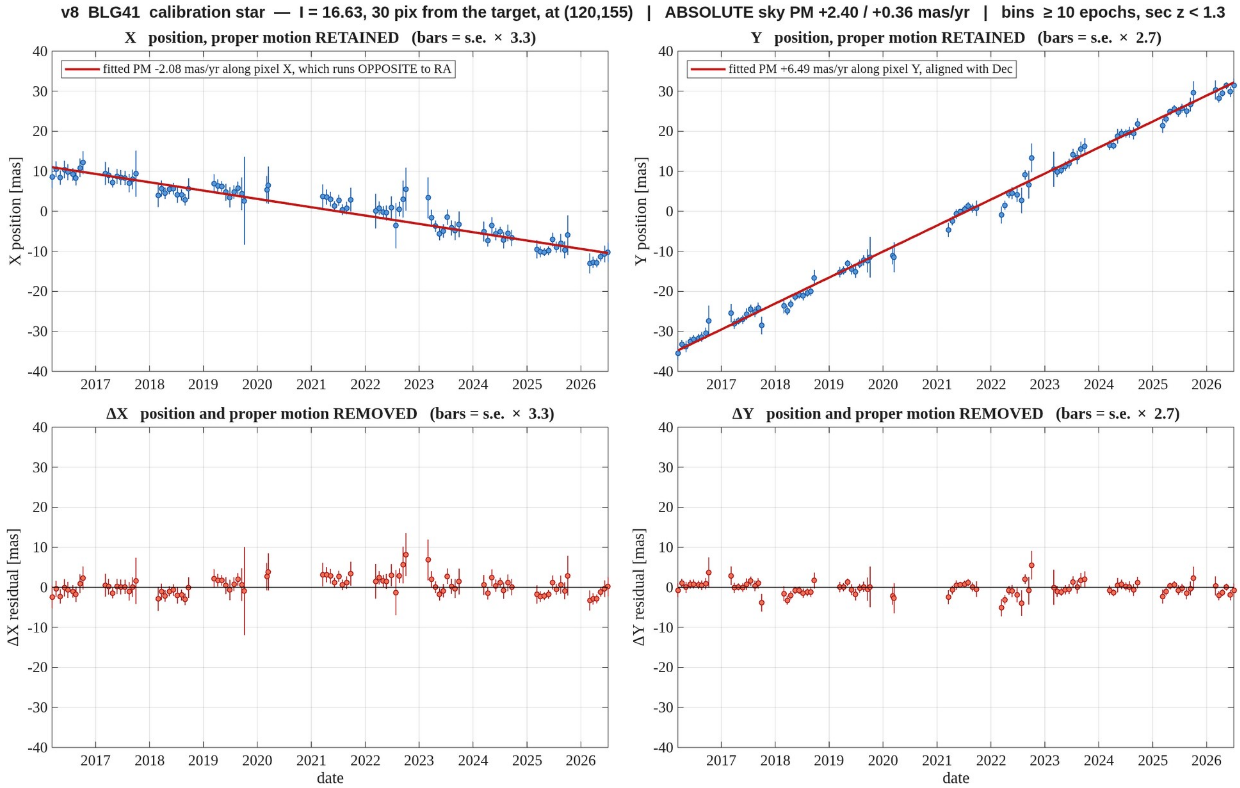


Figure 14. **BLG41** — I = 16.63, 30 pix away. Binned in one sidereal month, error bars are the standard error within the bin scaled by the empirical factor √(χ²/DoF) of the bin means (quoted in the panel titles, computed on the proper-motion-removed residuals and applied to both rows); bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.
`report_v8_motion_BLG41_m16_d30.png`

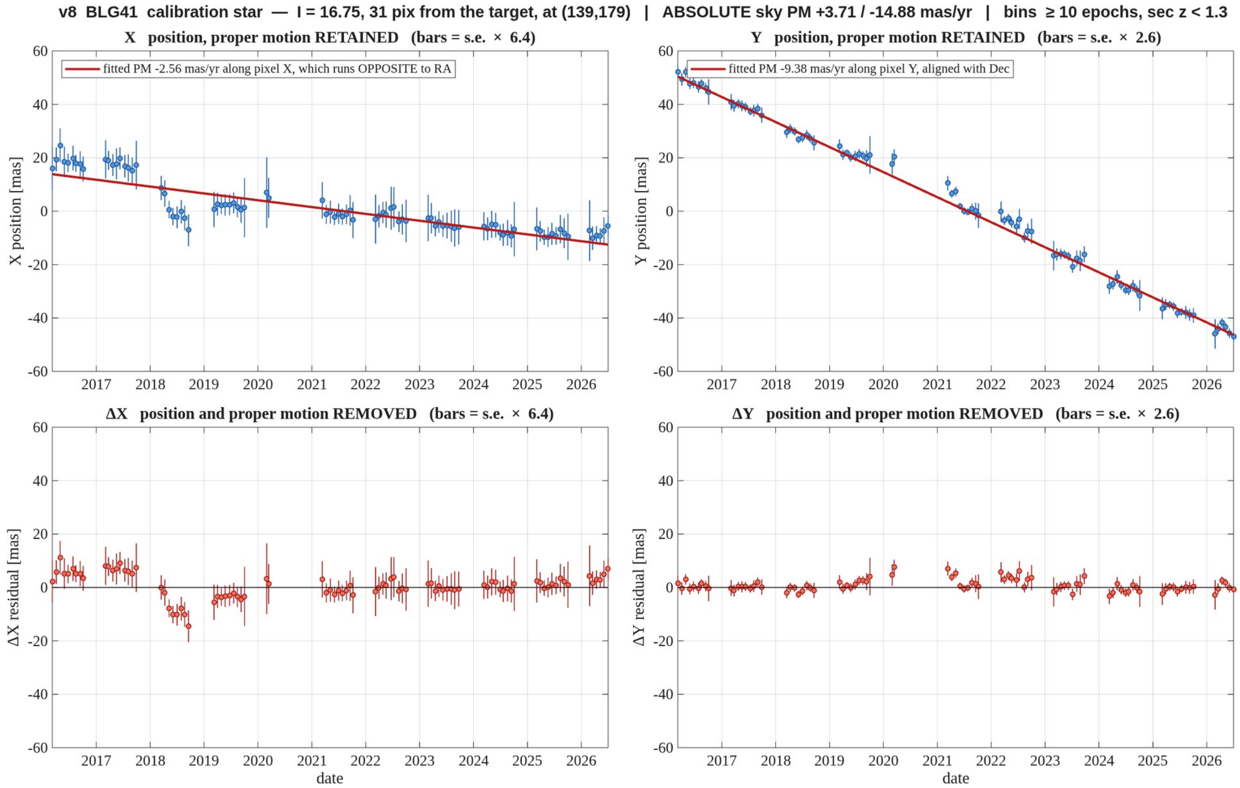


Figure 15. **BLG41** — I = 16.75, 31 pix away. Binned in one sidereal month, error bars are the standard error within the bin scaled by the empirical factor $\sqrt{(\chi^2/\mathrm{DoF})}$ of the bin means (quoted in the panel titles, computed on the proper-motion-removed residuals and applied to both rows); bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.
`report_v8_motion_BLG41_m16_d31.png`

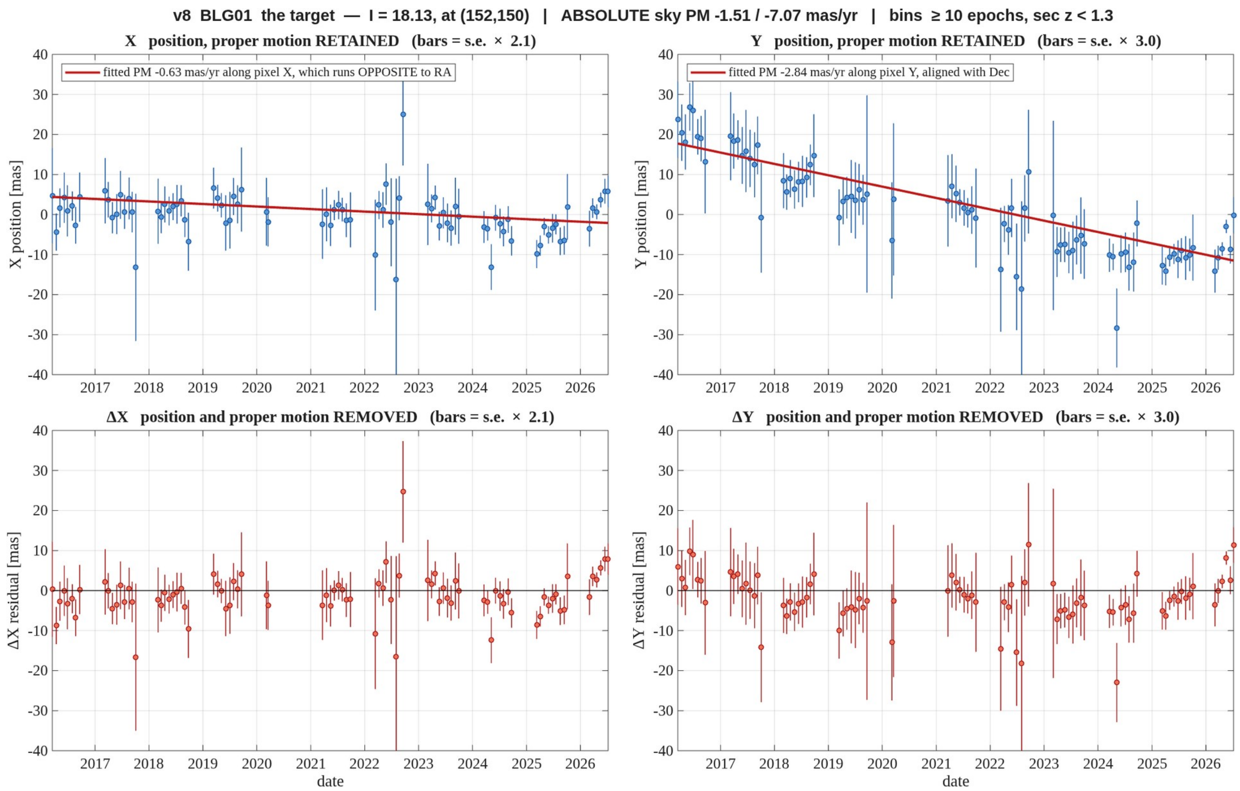


Figure 16. **BLG01** — the target, I = 18.13. Binned in one sidereal month, error bars are the standard error within the bin scaled by the empirical factor √(χ²/DoF) of the bin means (quoted in the panel titles, computed on the proper-motion-removed residuals and applied to both rows); bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v8_motion_BLG01_target.png`

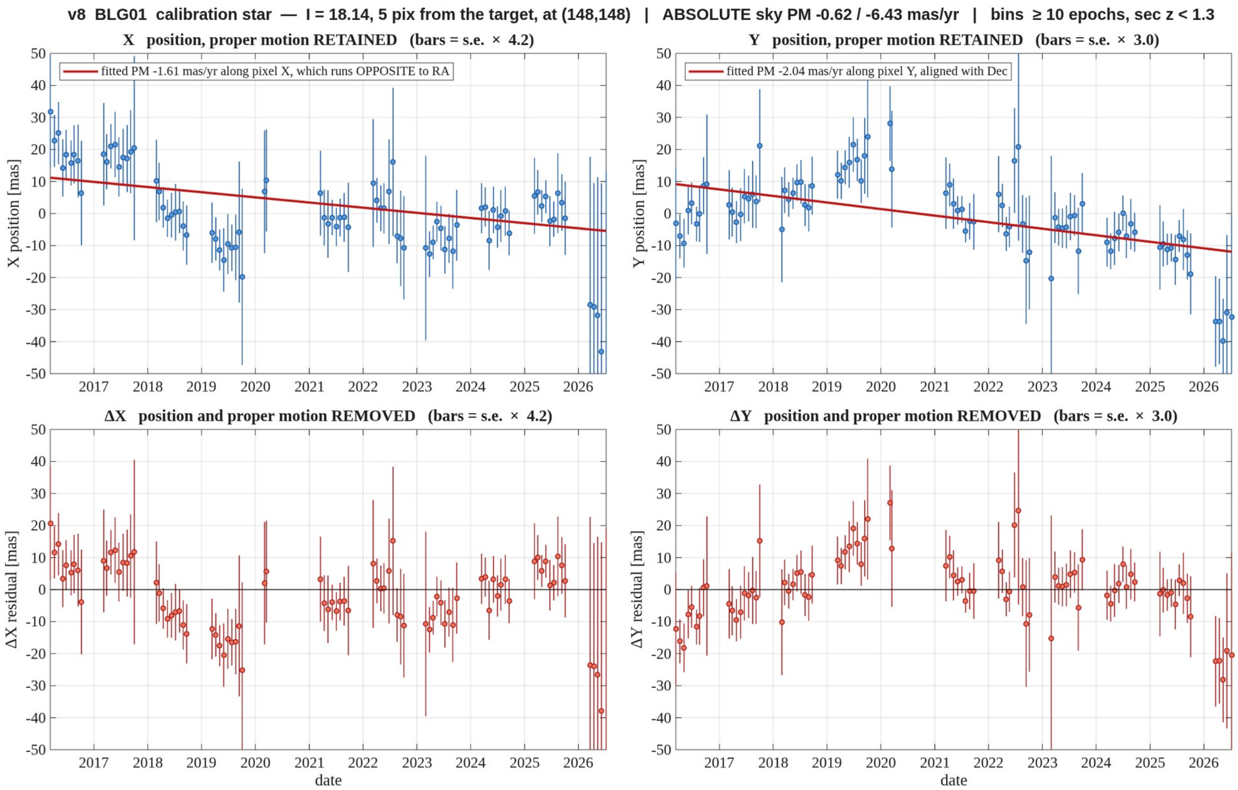


Figure 17. **BLG01** — I = 18.14, 4.5 pix away — **blended with the target**, see section 7. Binned in one sidereal month, error bars are the standard error within the bin scaled by the empirical factor √(χ²/DoF) of the bin means (quoted in the panel titles, computed on the proper-motion-removed residuals and applied to both rows); bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v8_motion_BLG01_m18_d04.png`

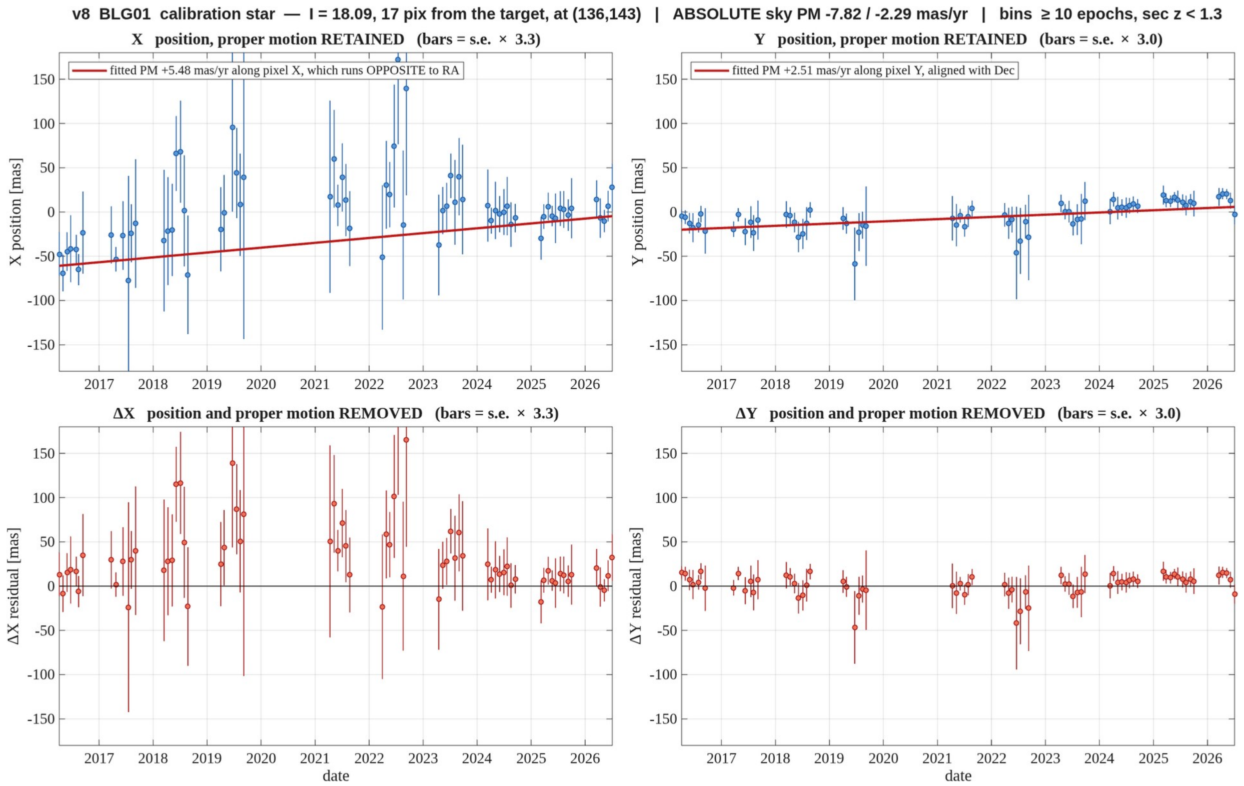


Figure 18. **BLG01** — I = 18.09, 17 pix away — poorly measured, see section 7. Binned in one sidereal month, error bars are the standard error within the bin scaled by the empirical factor √(χ²/DoF) of the bin means (quoted in the panel titles, computed on the proper-motion-removed residuals and applied to both rows); bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v8_motion_BLG01_m18_d17.png`

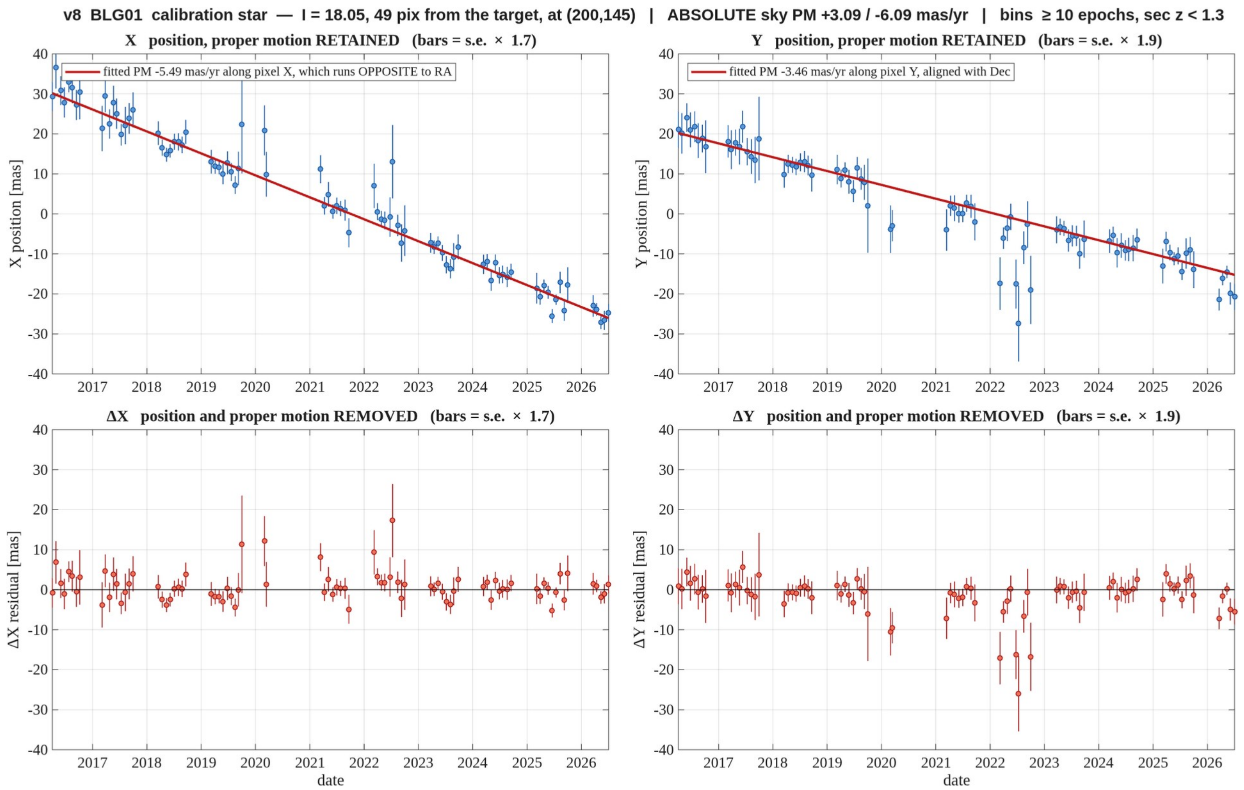


Figure 19. **BLG01** — I = 18.05, 48 pix away. Binned in one sidereal month, error bars are the standard error within the bin scaled by the empirical factor √(χ²/DoF) of the bin means (quoted in the panel titles, computed on the proper-motion-removed residuals and applied to both rows); bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.
`report_v8_motion_BLG01_m18_d48.png`

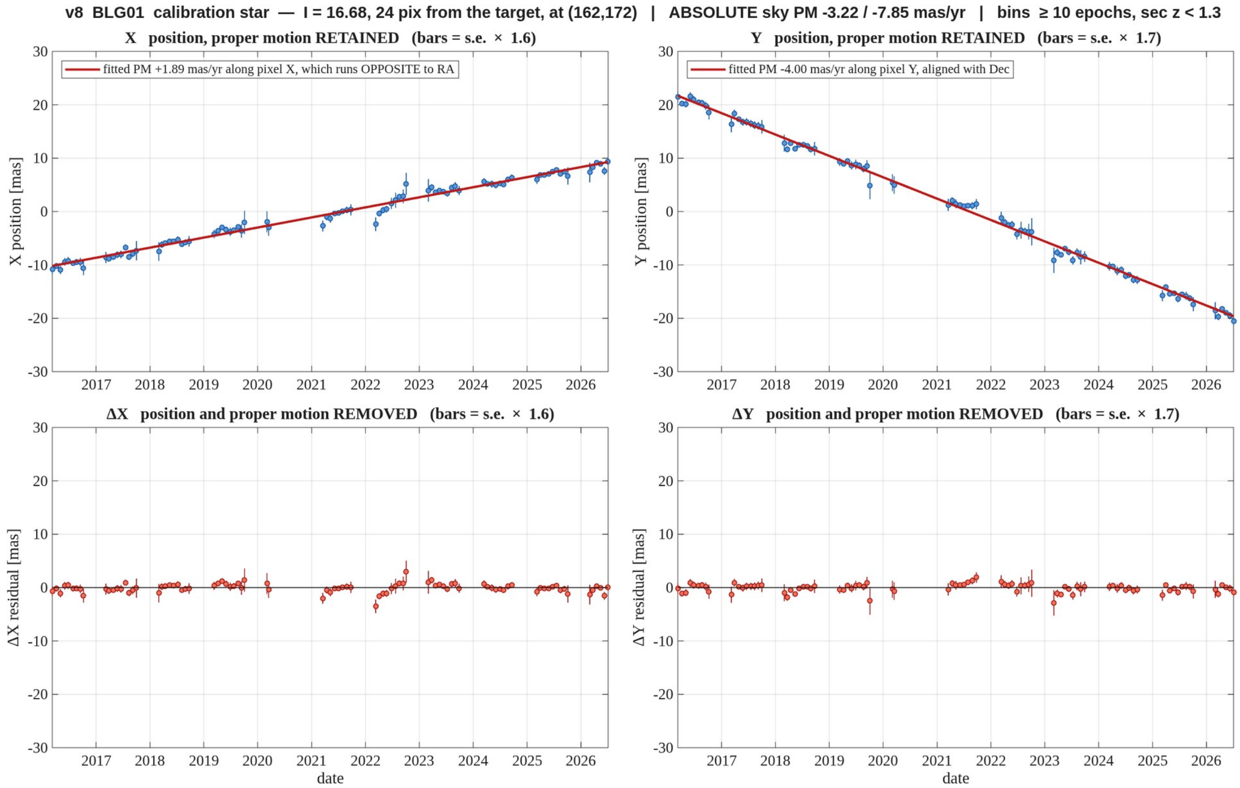


Figure 20. **BLG01** — I = 16.68, 24 pix away. Binned in one sidereal month, error bars are the standard error within the bin scaled by the empirical factor √(χ²/DoF) of the bin means (quoted in the panel titles, computed on the proper-motion-removed residuals and applied to both rows); bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.
`report_v8_motion_BLG01_m16_d24.png`

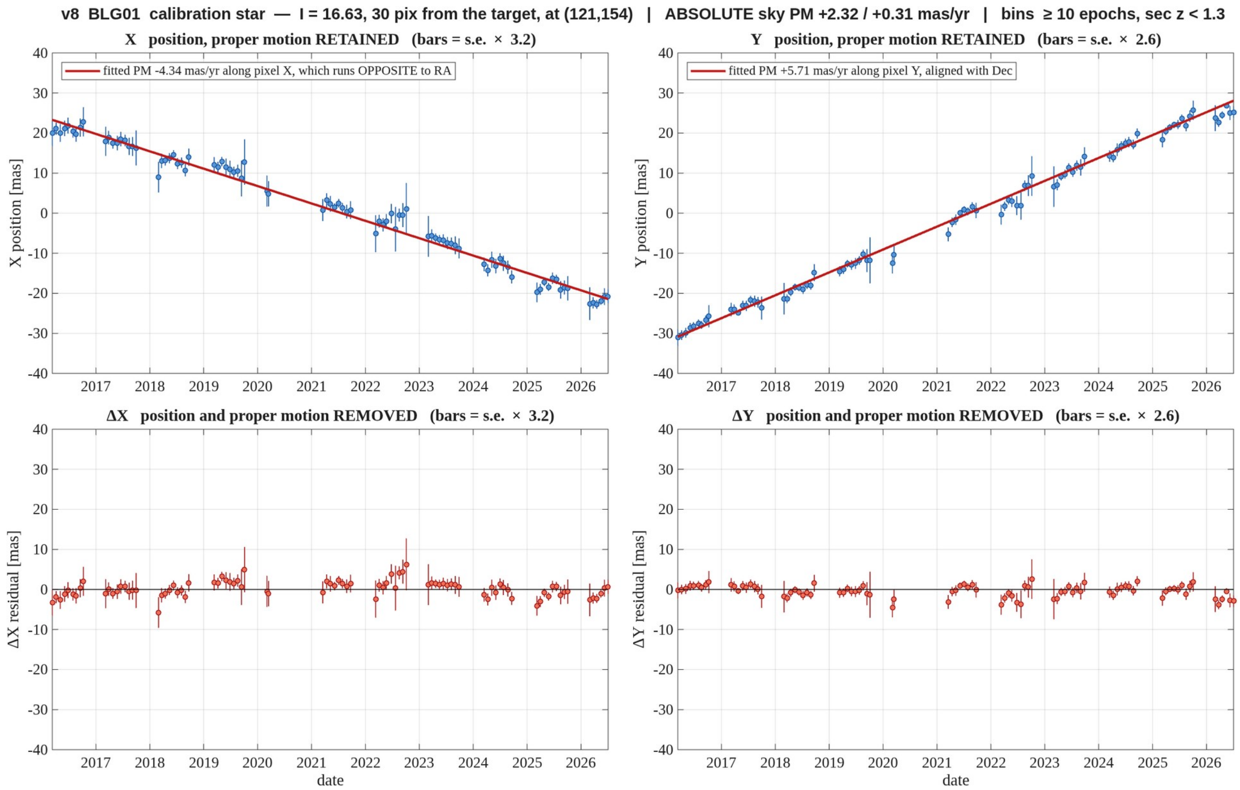


Figure 21. **BLG01** — I = 16.63, 30 pix away. Binned in one sidereal month, error bars are the standard error within the bin scaled by the empirical factor $\sqrt{(\chi^2/\mathrm{DoF})}$ of the bin means (quoted in the panel titles, computed on the proper-motion-removed residuals and applied to both rows); bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.
`report_v8_motion_BLG01_m16_d30.png`

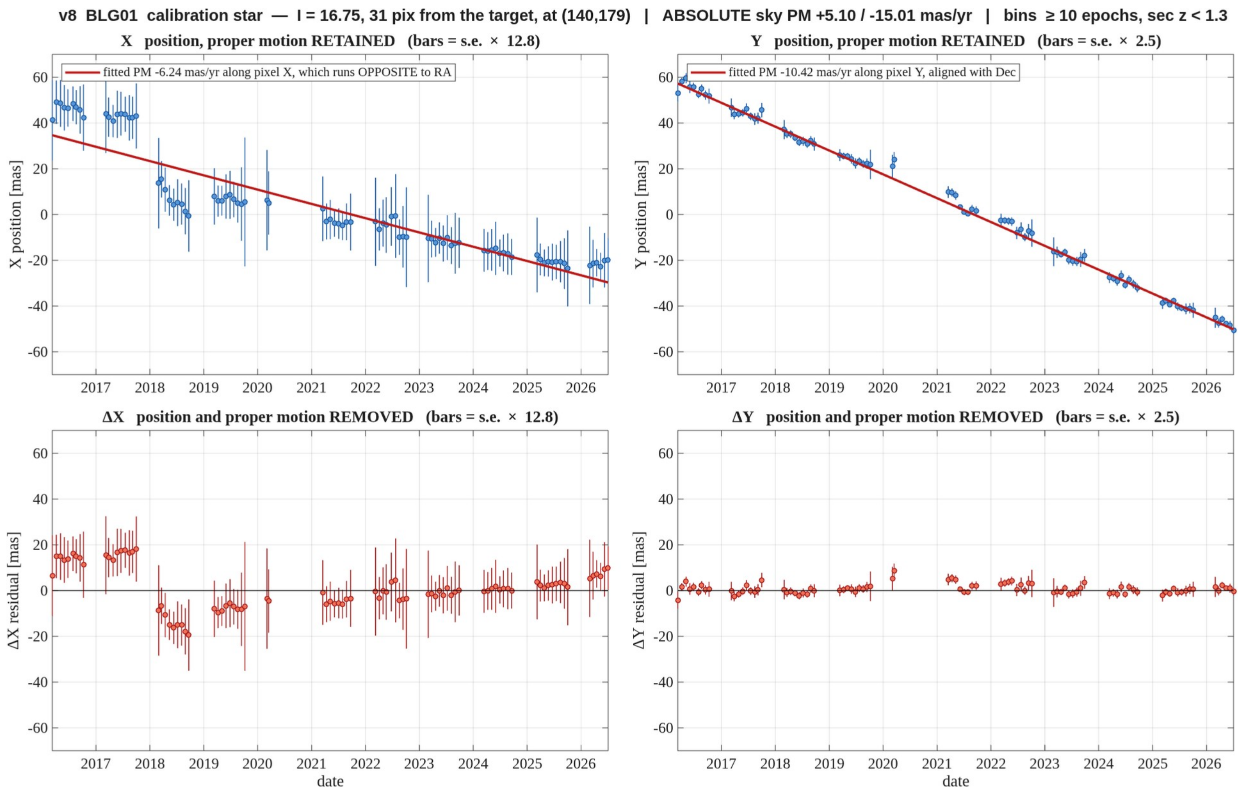


Figure 22. **BLG01** — I = 16.75, 31 pix away. Binned in one sidereal month, error bars are the standard error within the bin scaled by the empirical factor $\sqrt{\chi^2/\mathrm{DoF}}$ of the bin means (quoted in the panel titles, computed on the proper-motion-removed residuals and applied to both rows); bins with fewer than 10 epochs and epochs at sec z > 1.3 are dropped. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v8_motion_BLG01_m16_d31.png`